

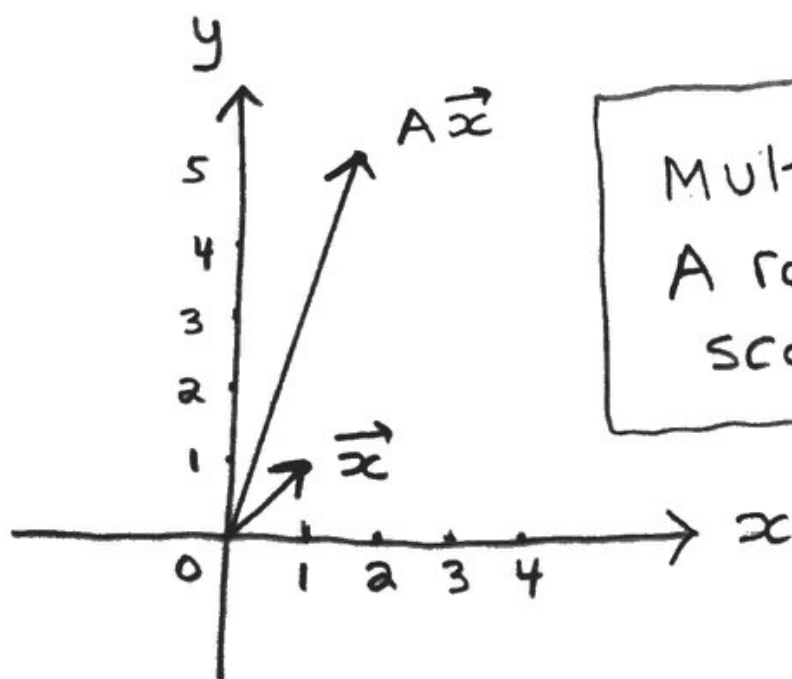
Chapter 4 Eigenvalues and Eigenvectors

4.1 Introduction to Eigenvalues and Eigenvectors

Suppose that A is a 2×2 matrix and $\vec{x}$ is a vector in $\mathbb{R}^2$. Typically, $A\vec{x}$ scales and rotates $\vec{x}$. For example,

if $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then

$$A\vec{x} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$



Multiplying $\vec{x}$ by A rotated and scaled it.

For certain vectors $\vec{k} \neq \vec{0}$, $A\vec{k}$ is parallel to $\vec{k}$, such a vector $\vec{k}$ is called an eigenvector of A .

Definition Let A be an $n \times n$ matrix.

A number λ is called an eigenvalue of A if there is a nonzero vector $\vec{k}$ such that :

$$A\vec{k} = \lambda \vec{k}$$

Such a vector $\vec{k}$ is called an eigenvector of A corresponding to λ .

ex: ① Show that $\vec{K}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and

$\vec{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$

and find their corresponding eigenvalues.

$$A\vec{K}_1 = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \underset{\substack{\uparrow \\ \text{eigenvalue}}}{2} \vec{K}_1$$

$$\boxed{\lambda_1 = 2}$$

$$A\vec{K}_2 = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \underset{\substack{\uparrow \\ \text{eigenvalue}}}{3} \vec{K}_2$$

$$\boxed{\lambda_2 = 3}$$

The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$.

Note: Even though $A\vec{0} = \lambda\vec{0}$ is always true, $\vec{0}$ is not an eigenvector.

Eigenvectors are nonzero vectors.

How can we find the eigenvalues of a square matrix?

Suppose that A is a 2×2 matrix. If λ is an eigenvalue of A , then:

$$A\vec{k} = \lambda\vec{k} \quad \text{for a vector } \vec{k} \neq \vec{0}$$

$$\Rightarrow A\vec{k} = \lambda I\vec{k} \quad (I = \text{identity matrix})$$

$$\Rightarrow A\vec{k} - \lambda I\vec{k} = \vec{0}$$

$$\Rightarrow (A - \lambda I)\vec{k} = \vec{0}$$

Since $\vec{k} \neq \vec{0}$ is a non trivial solution,
then:

$A - \lambda I$ is not invertible.

$$\Rightarrow \boxed{\det(A - \lambda I) = 0}$$

$$\det(A - \lambda I) = 0$$

is called the characteristic equation of A .

The solutions (or roots) of the equation are the eigenvalues of A .

ex: ② Find the eigenvalues of

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}.$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)(4-\lambda) + 2 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 6 = 0$$

$$\Rightarrow (\lambda - 2)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 2, 3 \text{ are the eigenvalues}$$

For a 2×2 matrix A ,

$$\det(A - \lambda I) = 0$$

is a quadratic equation.

A 2×2 matrix can have:

- ① Two distinct real eigenvalues
 - ② A single real eigenvalue
 - ③ Two complex conjugates eigenvalues
-

ex: ③ Find the eigenvalues of

$$A = \begin{bmatrix} -1 & -4 \\ 1 & -1 \end{bmatrix}.$$

$$A - \lambda I = \begin{bmatrix} -1-\lambda & -4 \\ 1 & -1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\Rightarrow (-1-\lambda)^2 + 4 = 0$$

$$\Rightarrow (-1-\lambda)^2 = -4$$

$$\Rightarrow -1-\lambda = \pm\sqrt{-4} = \pm 2i$$

$$\Rightarrow \boxed{\lambda = -1 \pm 2i}$$

there are two complex conjugates

eigenvalues: $\lambda_1 = -1 + 2i$ and $\lambda_2 = -1 - 2i$.

To find the eigenvectors corresponding to the eigenvalue λ , we solve:

$$(A - \lambda I) \vec{k} = \vec{0}.$$

We can do this by finding the RREF of

$$[A - \lambda I \mid \vec{0}].$$

The $\vec{0}$ column can be omitted since it remains $\vec{0}$ in the RREF.

Definition: The set of all eigenvectors corresponding to λ , plus the zero vector, is called the eigenspace of λ and is denoted by E_λ .

ex: ④ Find the eigenvalues and

eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$.

Eigenvalues: $A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{bmatrix}$

$$\det(A - \lambda I) = 0$$

$$\Rightarrow (1-\lambda)(4-\lambda) + 2 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 6 = 0$$

$$\Rightarrow (\lambda - 2)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 2, 3$$

The eigenvalues are $\lambda_1 = 2$

and $\lambda_2 = 3$.

Eigenvectors

We need to solve : $(A - \lambda I) \vec{k} = \vec{0}$

for each eigenvalue.

Assume : $\vec{k} = \begin{bmatrix} x \\ y \end{bmatrix}$.

For $\lambda_1 = 2$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix}$$

$$\left[A - \lambda_1 I \mid \vec{0} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x - 2y = 0 \rightarrow \begin{cases} x = 2t \\ y = t \end{cases}$$

$$\Rightarrow \vec{k} = \begin{bmatrix} 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ for } t \neq 0$$

are the eigenvectors for λ_1

$$\Rightarrow E_{\lambda_1} = \text{span} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right) \leftarrow \underline{\text{Eigenspace}}$$

For $\lambda_2 = 3$

$$A - \lambda_2 I = \begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix}$$

$$[A - \lambda_2 I | \vec{0}] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x - y = 0 \Rightarrow \begin{cases} x = t \\ y = t \end{cases}$$

$$\Rightarrow \vec{K} = \begin{bmatrix} t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for } t \neq 0$$

are the eigenvectors for λ_2 .

$$\Rightarrow E_{\lambda_2} = \text{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \leftarrow \text{Eigenspace}$$

Summary

To find the eigenvalues of A

Solve $\det(A - \lambda I) = 0$.

To find the eigenvectors $\vec{k} \neq \vec{0}$
corresponding to λ ,

Solve $(A - \lambda I) \vec{k} = \vec{0}$.
